

Element F

We used a matrix to combine our ideas and calculate which idea is the most viable and cost effective to conclude our rose mechanism idea that was used. To create our ideas, we researched other methods used to create the rose in Beauty and Beast and used that information to come up with five ideas each.

One of the biggest problems we encountered was time. We were on a time restraint and received materials late which caused our team to lose more time. Due to this problem, our team leader decided to use the motors from remote cars that were already set up allowing us to save time. Another benefit to using the remote motors from toy cars is our team already knew they worked. After talking to an engineering student at Duke, we used her suggestion of using an individual motor for each scene which allowed us to be more confident in our design. We utilized this idea by having one rose petal connected to individual motors via string.

Another issue we faced from materials delivering late was extra, unused time. In order to address this, we used the time to document, update elements and daily journals. After this, our extra time allowed for team building which significantly increased our team communication.

Overall our design will fulfill our design requirements. It is controlled via remote, so the rose will stay on the stand for the duration of the play. Because we used parts from a car, we know that they already work, reducing the risk of hazard of crew members and decreasing the risk of mechanical or operational issues. It will also come with the battery packs so once the batteries are put in, it will have access to the amount of volts required. Our visual design of the flower will allow it to be 1.5 feet tall, have three red petals and a green stem.